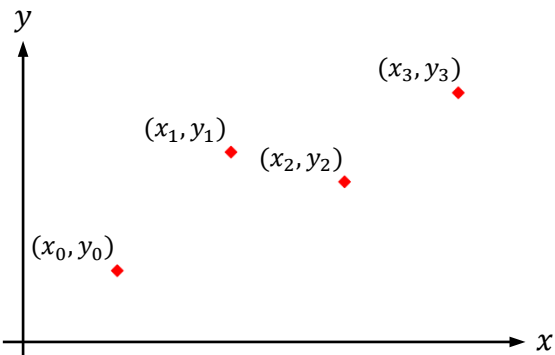
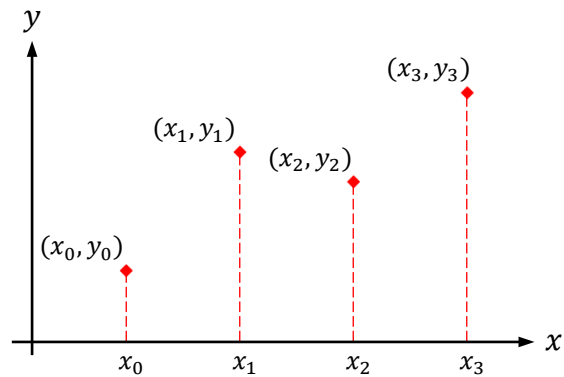


The Method of Undetermined Coefficients

The Method of Undetermined Coefficients

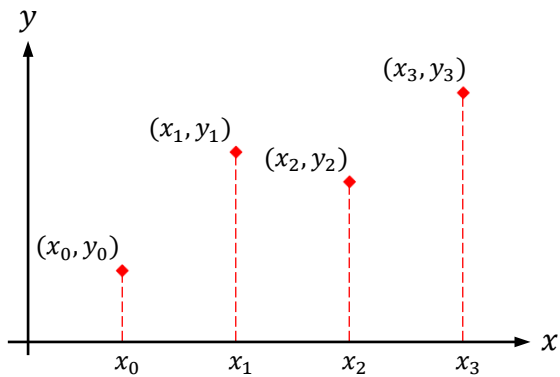


The Method of Undetermined Coefficients



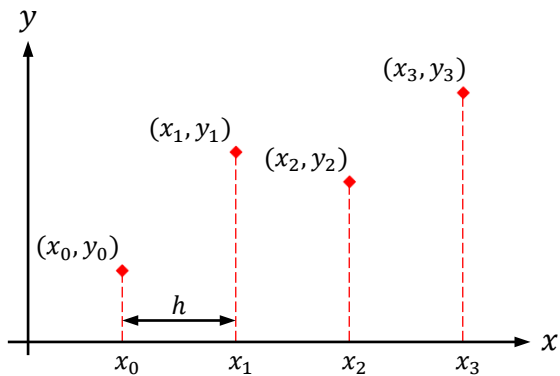
The Method of Undetermined Coefficients

Formulas for Uniform Intervals



The Method of Undetermined Coefficients

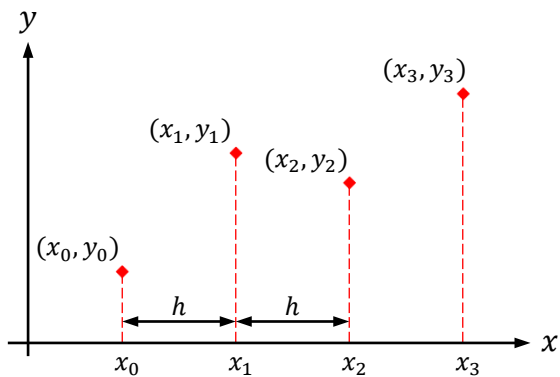
Formulas for Uniform Intervals



$$x_1 - x_0 = h$$

The Method of Undetermined Coefficients

Formulas for Uniform Intervals

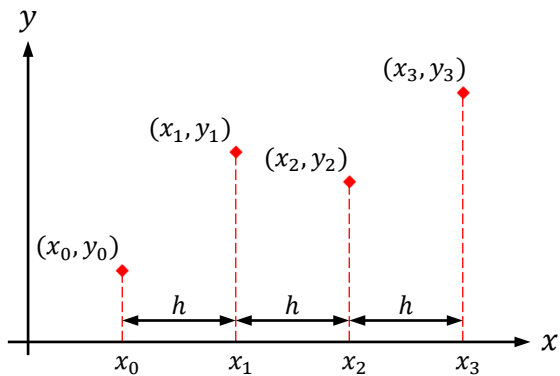


$$x_1 - x_0 = h$$

$$x_2 - x_1 = h$$

The Method of Undetermined Coefficients

Formulas for Uniform Intervals



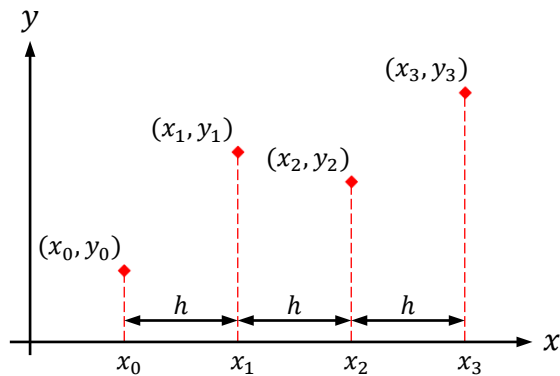
$$x_1 - x_0 = h$$

$$x_2 - x_1 = h$$

$$x_3 - x_2 = h$$

The Method of Undetermined Coefficients

Formulas for Uniform Intervals



$$x_1 - x_0 = h$$

$$x_2 - x_1 = h$$

$$x_3 - x_2 = h$$

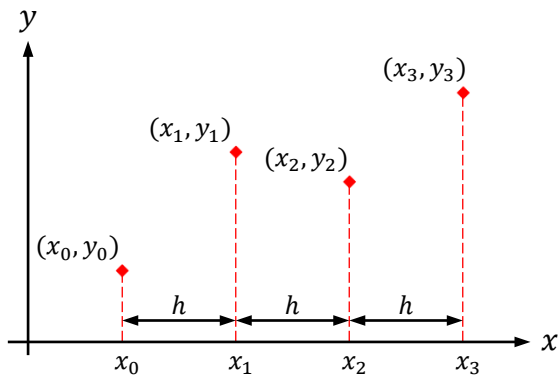
$$\vdots$$

$$x_{i+1} - x_i = h$$

The Method of Undetermined Coefficients

Formulas for Uniform Intervals

If data at $n + 1$ points is provided, $(x_i, f(x_i)) \mid i = 0, 1, 2, \dots, n$; k^{th} derivative of $f(x)$ at x_i when $1 \leq k \leq n$ can be approximated by the following expression.



$$x_1 - x_0 = h$$

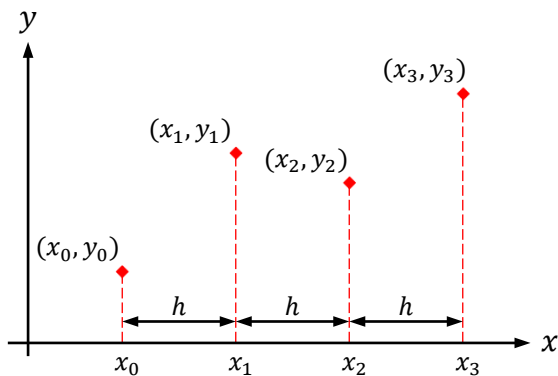
$$x_2 - x_1 = h$$

$$x_3 - x_2 = h$$

$$\vdots$$

$$x_{i+1} - x_i = h$$

The Method of Undetermined Coefficients



$$x_1 - x_0 = h$$

$$x_2 - x_1 = h$$

$$x_3 - x_2 = h$$

$$\vdots$$

$$x_{i+1} - x_i = h$$

Formulas for Uniform Intervals

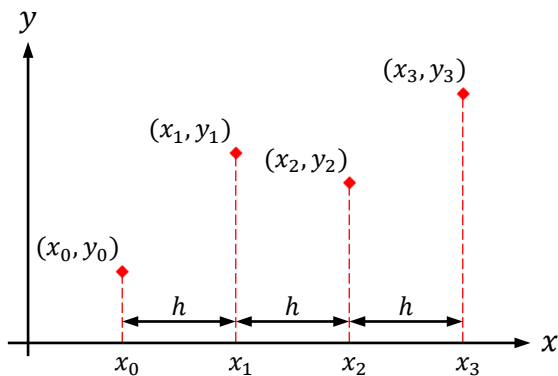
If data at $n + 1$ points is provided, $(x_i, f(x_i)) \mid i = 0, 1, 2, \dots, n$; k^{th} derivative of $f(x)$ at x_i when $1 \leq k \leq n$ can be approximated by the following expression.

$$\sum_{j=m}^{m+n} w_{j-m} \cdot f(x_{i+j})$$

$$= w_0 \cdot f(x_{i+m}) + w_1 \cdot f(x_{i+m+1}) + \dots + w_n \cdot f(x_{i+m+n}); \quad m \leq 0 < m+n, n \geq 1$$

The Method of Undetermined Coefficients

Formulas for Uniform Intervals



$$x_1 - x_0 = h$$

$$x_2 - x_1 = h$$

$$x_3 - x_2 = h$$

$$\vdots$$

$$x_{i+1} - x_i = h$$

If data at $n + 1$ points is provided, $(x_i, f(x_i)) \mid i = 0, 1, 2, \dots, n$; k^{th} derivative of $f(x)$ at x_i when $1 \leq k \leq n$ can be approximated by the following expression.

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Taylor series expanded around x_i are substituted into the equation, then coefficients of the first $n + 1$ terms, including k^{th} derivative, are collected to set up system of equations to find all weighting factors.

The Method of Undetermined Coefficients

The Method of Undetermined Coefficients

Forward Difference Formulas

The Method of Undetermined Coefficients

Forward Difference Formulas

$$m = 0$$

The Method of Undetermined Coefficients

Forward Difference Formulas

$$m = 0 \quad \sum_{j=0}^n w_j \cdot f(x_{i+j}) = w_0 \cdot f(x_i) + w_1 \cdot f(x_{i+1}) + \cdots + w_n \cdot f(x_{i+n})$$

The Method of Undetermined Coefficients

Forward Difference Formulas

 $m = 0$

$$\sum_{j=0}^n w_j \cdot f(x_{i+j}) = w_0 \cdot f(x_i) + w_1 \cdot f(x_{i+1}) + \cdots + w_n \cdot f(x_{i+n})$$

$$= w_0 \cdot f(x_i)$$

$$+ w_1 \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \cdots \right]$$

$$+ w_2 \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \cdots \right]$$

$$\vdots$$

$$+ w_n \cdot \left[f(x_i) + (nh) \cdot f'(x_i) + \frac{(nh)^2}{2!} \cdot f''(x_i) + \cdots \right]$$

The Method of Undetermined Coefficients

Forward Difference Formulas

$$m = 0$$

$$\sum_{j=0}^n w_j \cdot f(x_{i+j}) = w_0 \cdot f(x_i) + w_1 \cdot f(x_{i+1}) + \cdots + w_n \cdot f(x_{i+n})$$

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$$\vdots$$

$$+ w_n \cdot \left[f(x_i) + (nh) \cdot f'(x_i) + \frac{(nh)^2}{2!} \cdot f''(x_i) + \cdots \right]$$

Setting Up System of Equations.

Coefficient of $f^{(k)}(x_i)$ is equal to 1.

$$w_0 + w_1 + w_2 + \cdots + w_n = 0$$

$$w_1 \cdot (h) + w_2 \cdot (2h) + \cdots + w_n \cdot (nh) = 0$$

$$\vdots$$

$$w_1 \cdot \frac{(h)^k}{k!} + w_2 \cdot \frac{(2h)^k}{k!} + \cdots + w_n \cdot \frac{(nh)^k}{k!} = 1$$

$$\vdots$$

$$w_1 \cdot \frac{(h)^n}{n!} + w_2 \cdot \frac{(2h)^n}{n!} + \cdots + w_n \cdot \frac{(nh)^n}{n!} = 0$$

The Method of Undetermined Coefficients

Example 05-01: Find the formula for $f'(x_i)$ using the values of $f(x_i), f(x_{i+1}), f(x_{i+2})$.

The Method of Undetermined Coefficients

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$$a \cdot f(x_i) + b \cdot f(x_{i+1}) + c \cdot f(x_{i+2})$$

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Coefficient of $f'(x_i)$ is equal to 1.

$$a + b + c = 0$$

$$b \cdot (h) + c \cdot (2h) = 1$$

$$b \cdot \frac{(h)^2}{2!} + c \cdot \frac{(2h)^2}{2!} = 0$$

The Method of Undetermined Coefficients

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$$a = -\frac{3}{2h}, b = \frac{2}{h}, c = -\frac{1}{2h}$$

The Method of Undetermined Coefficients

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$$\frac{-3 \cdot f(x_i) + 4 \cdot f(x_{i+1}) - f(x_{i+2})}{2h} = f'(x_i) + \left[\left(\frac{2}{h} \right) \cdot \frac{(h)^3}{3!} + \left(-\frac{1}{2h} \right) \cdot \frac{(2h)^3}{3!} \right] \cdot f^{(3)}(x_i) + \dots$$

The Method of Undetermined Coefficients

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$$\frac{-3 \cdot f(x_i) + 4 \cdot f(x_{i+1}) - f(x_{i+2})}{2h} = f'(x_i) - \frac{h^2}{3} \cdot f^{(3)}(x_i) + \dots$$

Coefficient of $f'(x_i)$ is equal to 1.

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$$f'(x_i) = \frac{-3 \cdot f(x_i) + 4 \cdot f(x_{i+1}) - f(x_{i+2})}{2h} + \frac{h^2}{3} \cdot f^{(3)}(x_i) + \dots$$

The Method of Undetermined Coefficients

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$$f'(x_i) = \frac{-3 \cdot f(x_i) + 4 \cdot f(x_{i+1}) - f(x_{i+2})}{2h} + \frac{h^2}{3} \cdot f^{(3)}(x_i) + \dots$$

$$f'(x_i) \approx \frac{-3 \cdot f(x_i) + 4 \cdot f(x_{i+1}) - f(x_{i+2})}{2h}$$

The Method of Undetermined Coefficients

Example 05-01: Find the formula for $f'(x_i)$ using the values of $f(x_i), f(x_{i+1}), f(x_{i+2})$.

$$a \cdot f(x_i) + b \cdot f(x_{i+1}) + c \cdot f(x_{i+2})$$

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$$f'(x_i) = \frac{-3 \cdot f(x_i) + 4 \cdot f(x_{i+1}) - f(x_{i+2})}{2h} + \frac{h^2}{3} \cdot f^{(3)}(x_i) + \dots$$

$$f'(x_i) \approx \frac{-3 \cdot f(x_i) + 4 \cdot f(x_{i+1}) - f(x_{i+2})}{2h}$$

$$\varepsilon = \frac{h^2}{3} \cdot f^{(3)}(\xi); \quad x_i \leq \xi \leq x_{i+2}$$

The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

x_i	$f(x_i)$
0.5	-4.827866
0.6	-4.240058
0.7	-3.496909
0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

x_i	$f(x_i)$
0.5	-4.827866
0.6	-4.240058
0.7	-3.496909
0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

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The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

	x_i	$f(x_i)$
i	0.5	-4.827866
$i + 1$	0.6	-4.240058
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$i + 3$	0.8	-2.596792

$$a \cdot f(x_i) + b \cdot f(x_{i+1}) + c \cdot f(x_{i+2}) + d \cdot f(x_{i+3})$$

The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

	x_i	$f(x_i)$
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$i + 2$	0.7	-3.496909
$i + 3$	0.8	-2.596792

$$\begin{aligned} & a \cdot f(x_i) + b \cdot f(x_{i+1}) + c \cdot f(x_{i+2}) + d \cdot f(x_{i+3}) \\ &= a \cdot f(x_i) \\ &+ b \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ c \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ d \cdot \left[f(x_i) + (3h) \cdot f'(x_i) + \frac{(3h)^2}{2!} \cdot f''(x_i) + \frac{(3h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \end{aligned}$$

The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

	x_i	$f(x_i)$
i	0.5	-4.827866
$i + 1$	0.6	-4.240058
$i + 2$	0.7	-3.496909
$i + 3$	0.8	-2.596792

$$a \cdot f(x_i) + b \cdot f(x_{i+1}) + c \cdot f(x_{i+2}) + d \cdot f(x_{i+3})$$

$$= a \cdot f(x_i)$$

$$+ b \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ c \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ d \cdot \left[f(x_i) + (3h) \cdot f'(x_i) + \frac{(3h)^2}{2!} \cdot f''(x_i) + \frac{(3h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & (1)^2 & (2)^2 & (3)^2 \\ 0 & (1)^3 & (2)^3 & (3)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

	x_i	$f(x_i)$
i	0.5	-4.827866
$i + 1$	0.6	-4.240058
$i + 2$	0.7	-3.496909
$i + 3$	0.8	-2.596792

$$a \cdot f(x_i) + b \cdot f(x_{i+1}) + c \cdot f(x_{i+2}) + d \cdot f(x_{i+3})$$

$$= a \cdot f(x_i)$$

$$+ b \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ c \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ d \cdot \left[f(x_i) + (3h) \cdot f'(x_i) + \frac{(3h)^2}{2!} \cdot f''(x_i) + \frac{(3h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & (1)^2 & (2)^2 & (3)^2 \\ 0 & (1)^3 & (2)^3 & (3)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

$$a = -\frac{11}{6h}, b = \frac{3}{h}, c = -\frac{3}{2h}, d = \frac{1}{3h}$$

The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

	x_i	$f(x_i)$
i	0.5	-4.827866
$i + 1$	0.6	-4.240058
$i + 2$	0.7	-3.496909
$i + 3$	0.8	-2.596792

$$a \cdot f(x_i) + b \cdot f(x_{i+1}) + c \cdot f(x_{i+2}) + d \cdot f(x_{i+3})$$

$$= a \cdot f(x_i)$$

$$+ b \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ c \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ d \cdot \left[f(x_i) + (3h) \cdot f'(x_i) + \frac{(3h)^2}{2!} \cdot f''(x_i) + \frac{(3h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & (1)^2 & (2)^2 & (3)^2 \\ 0 & (1)^3 & (2)^3 & (3)^3 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{pmatrix}$$

$$f'(x_i) \approx \frac{-11 \cdot f(x_i) + 18 \cdot f(x_{i+1}) - 9 \cdot f(x_{i+2}) + 2 \cdot f(x_{i+3})}{6h}$$

$$a = -\frac{11}{6h}, b = \frac{3}{h}, c = -\frac{3}{2h}, d = \frac{1}{3h}$$

The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

	x_i	$f(x_i)$
i	0.5	-4.827866
$i + 1$	0.6	-4.240058
$i + 2$	0.7	-3.496909
$i + 3$	0.8	-2.596792

$$\begin{aligned} & a \cdot f(x_i) + b \cdot f(x_{i+1}) + c \cdot f(x_{i+2}) + d \cdot f(x_{i+3}) \\ &= a \cdot f(x_i) \\ &+ b \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ c \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ d \cdot \left[f(x_i) + (3h) \cdot f'(x_i) + \frac{(3h)^2}{2!} \cdot f''(x_i) + \frac{(3h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \end{aligned}$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & (1)^2 & (2)^2 & (3)^2 \\ 0 & (1)^3 & (2)^3 & (3)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

$$a = -\frac{11}{6h}, b = \frac{3}{h}, c = -\frac{3}{2h}, d = \frac{1}{3h}$$

$$\begin{aligned} f'(x_i) &\approx \frac{-11 \cdot f(x_i) + 18 \cdot f(x_{i+1}) - 9 \cdot f(x_{i+2}) + 2 \cdot f(x_{i+3})}{6h} \\ f'(0.5) &\approx \frac{-11 \cdot (-4.827866) + 18 \cdot (-4.240058) - 9 \cdot (-3.496909) + 2 \cdot (-2.596792)}{6 \cdot (0.1)} \end{aligned}$$

The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

	x_i	$f(x_i)$
i	0.5	-4.827866
$i + 1$	0.6	-4.240058
$i + 2$	0.7	-3.496909
$i + 3$	0.8	-2.596792

$$a \cdot f(x_i) + b \cdot f(x_{i+1}) + c \cdot f(x_{i+2}) + d \cdot f(x_{i+3})$$

$$= a \cdot f(x_i)$$

$$+ b \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ c \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ d \cdot \left[f(x_i) + (3h) \cdot f'(x_i) + \frac{(3h)^2}{2!} \cdot f''(x_i) + \frac{(3h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & (1)^2 & (2)^2 & (3)^2 \\ 0 & (1)^3 & (2)^3 & (3)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

$$a = -\frac{11}{6h}, b = \frac{3}{h}, c = -\frac{3}{2h}, d = \frac{1}{3h}$$

$$f'(x_i) \approx \frac{-11 \cdot f(x_i) + 18 \cdot f(x_{i+1}) - 9 \cdot f(x_{i+2}) + 2 \cdot f(x_{i+3})}{6h}$$

$$f'(0.5) \approx \frac{-11 \cdot (-4.827866) + 18 \cdot (-4.240058) - 9 \cdot (-3.496909) + 2 \cdot (-2.596792)}{6 \cdot (0.1)}$$

$$= 5.106798$$

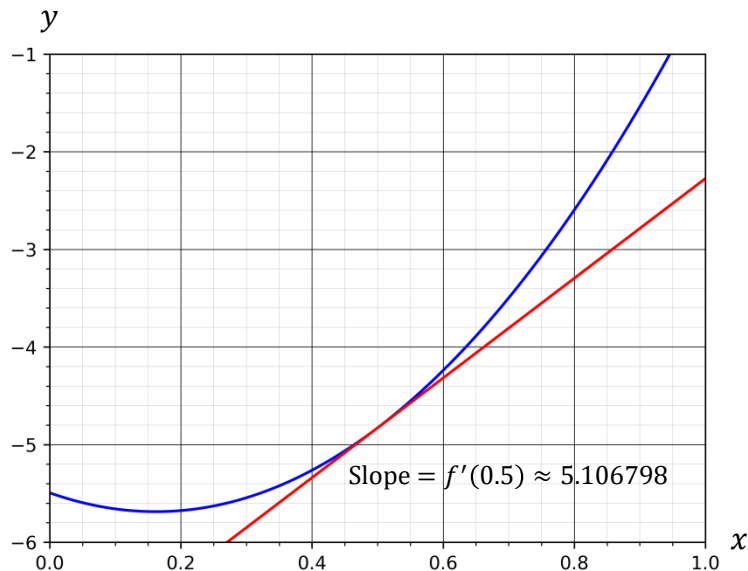
The Method of Undetermined Coefficients

Example 05-02: Approximate the value of $f'(0.5)$ from the following data.

	x_i	$f(x_i)$
i	0.5	-4.827866
$i + 1$	0.6	-4.240058
$i + 2$	0.7	-3.496909
$i + 3$	0.8	-2.596792

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & (1)^2 & (2)^2 & (3)^2 \\ 0 & (1)^3 & (2)^3 & (3)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

$$a = -\frac{11}{6h}, b = \frac{3}{h}, c = -\frac{3}{2h}, d = \frac{1}{3h}$$



The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

x_i	$f(x_i)$
0.5	-4.827866
0.6	-4.240058
0.7	-3.496909
0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

x_i	$f(x_i)$
0.5	-4.827866
0.6	-4.240058
0.7	-3.496909
0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
	0.5	-4.827866
<i>i</i>	0.6	-4.240058
	0.7	-3.496909
	0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2})$$

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
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i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$\begin{aligned}
 & a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2}) \\
 &= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ b \cdot f(x_i) \\
 &+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]
 \end{aligned}$$

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
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$i + 2$	0.8	-2.596792

$$\begin{aligned}
 & a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2}) \\
 &= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ b \cdot f(x_i) \\
 &+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]
 \end{aligned}$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
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i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$\begin{aligned}
 & \mathbf{a} \cdot f(x_{i-1}) + \mathbf{b} \cdot f(x_i) + \mathbf{c} \cdot f(x_{i+1}) + \mathbf{d} \cdot f(x_{i+2}) \\
 &= \mathbf{a} \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ \mathbf{b} \cdot f(x_i) \\
 &+ \mathbf{c} \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ \mathbf{d} \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]
 \end{aligned}$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

$$\mathbf{a} = -\frac{1}{3h}, \mathbf{b} = -\frac{1}{2h}, \mathbf{c} = \frac{1}{h}, \mathbf{d} = -\frac{1}{6h}$$

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
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i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$\begin{aligned}
 & a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2}) \\
 &= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ b \cdot f(x_i) \\
 &+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]
 \end{aligned}$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{pmatrix}$$

$$f'(x_i) \approx \frac{-2 \cdot f(x_{i-1}) - 3 \cdot f(x_i) + 6 \cdot f(x_{i+1}) - f(x_{i+2}))}{6h}$$

$$a = -\frac{1}{3h}, b = -\frac{1}{2h}, c = \frac{1}{h}, d = -\frac{1}{6h}$$

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
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$i + 2$	0.8	-2.596792

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

$$a = -\frac{1}{3h}, b = -\frac{1}{2h}, c = \frac{1}{h}, d = -\frac{1}{6h}$$

$$\begin{aligned} & a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2}) \\ &= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ b \cdot f(x_i) \\ &+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \end{aligned}$$

$$f'(x_i) \approx \frac{-2 \cdot f(x_{i-1}) - 3 \cdot f(x_i) + 6 \cdot f(x_{i+1}) - f(x_{i+2}))}{6h}$$

$$f'(0.6) \approx \frac{-2 \cdot (-4.827866) - 3 \cdot (-4.240058) + 6 \cdot (-3.496909) - (-2.596792)}{6 \cdot (0.1)}$$

The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

$$a = -\frac{1}{3h}, b = -\frac{1}{2h}, c = \frac{1}{h}, d = -\frac{1}{6h}$$

$$\begin{aligned} & a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2}) \\ &= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ b \cdot f(x_i) \\ &+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \end{aligned}$$

$$f'(x_i) \approx \frac{-2 \cdot f(x_{i-1}) - 3 \cdot f(x_i) + 6 \cdot f(x_{i+1}) - f(x_{i+2}))}{6h}$$

$$f'(0.6) \approx \frac{-2 \cdot (-4.827866) - 3 \cdot (-4.240058) + 6 \cdot (-3.496909) - (-2.596792)}{6 \cdot (0.1)}$$

$$= 6.652073$$

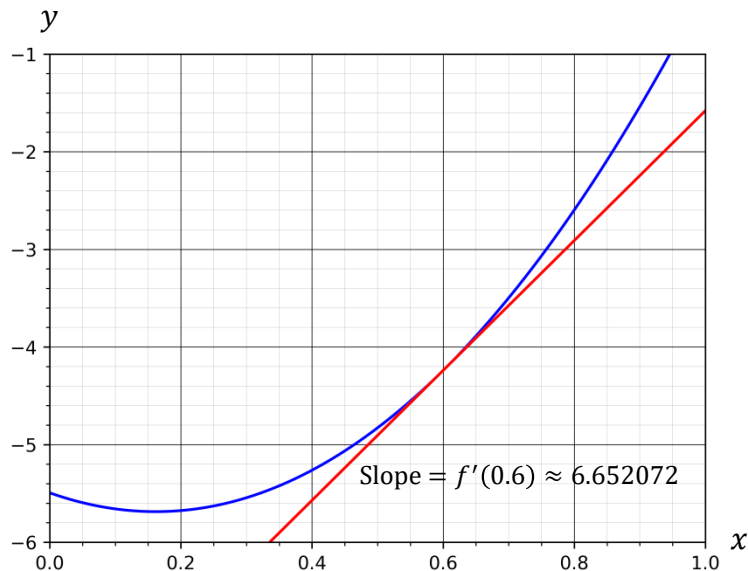
The Method of Undetermined Coefficients

Example 05-03: Approximate the value of $f'(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ \frac{1}{h} \\ 0 \\ 0 \end{Bmatrix}$$

$$a = -\frac{1}{3h}, b = -\frac{1}{2h}, c = \frac{1}{h}, d = -\frac{1}{6h}$$



The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

x_i	$f(x_i)$
0.5	-4.827866
0.6	-4.240058
0.7	-3.496909
0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

x_i	$f(x_i)$
0.5	-4.827866
0.6	-4.240058
0.7	-3.496909
0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

	x_i	$f(x_i)$
	0.5	-4.827866
<i>i</i>	0.6	-4.240058
	0.7	-3.496909
	0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2})$$

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$\begin{aligned} & a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2}) \\ &= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ b \cdot f(x_i) \\ &+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\ &+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \end{aligned}$$

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$\begin{aligned}
 & a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2}) \\
 &= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ b \cdot f(x_i) \\
 &+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]
 \end{aligned}$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{6}{h^3} \end{pmatrix}$$

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$\begin{aligned}
 & a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2}) \\
 &= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ b \cdot f(x_i) \\
 &+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]
 \end{aligned}$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{6}{h^3} \end{pmatrix}$$

$$a = -\frac{1}{h^3}, b = \frac{3}{h^3}, c = -\frac{3}{h^3}, d = \frac{1}{h^3}$$

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$\begin{aligned}
 & a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2}) \\
 &= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ b \cdot f(x_i) \\
 &+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right] \\
 &+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]
 \end{aligned}$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{6}{h^3} \end{pmatrix}$$

$$f^{(3)}(x_i) \approx \frac{-f(x_{i-1}) + 3 \cdot f(x_i) - 3 \cdot f(x_{i+1}) + f(x_{i+2})}{h^3}$$

$$a = -\frac{1}{h^3}, b = \frac{3}{h^3}, c = -\frac{3}{h^3}, d = \frac{1}{h^3}$$

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2})$$

$$= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ b \cdot f(x_i)$$

$$+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{6}{h^3} \end{pmatrix}$$

$$f^{(3)}(x_i) \approx \frac{-f(x_{i-1}) + 3 \cdot f(x_i) - 3 \cdot f(x_{i+1}) + f(x_{i+2})}{h^3}$$

$$f^{(3)}(0.6) \approx \frac{-(-4.827866) + 3 \cdot (-4.240058) - 3 \cdot (-3.496909) + (-2.596792)}{(0.1)^3}$$

$$a = -\frac{1}{h^3}, b = \frac{3}{h^3}, c = -\frac{3}{h^3}, d = \frac{1}{h^3}$$

The Method of Undetermined Coefficients

Example 05-04: Approximate the value of $f^{(3)}(0.6)$ from the following data.

	x_i	$f(x_i)$
$i - 1$	0.5	-4.827866
i	0.6	-4.240058
$i + 1$	0.7	-3.496909
$i + 2$	0.8	-2.596792

$$a \cdot f(x_{i-1}) + b \cdot f(x_i) + c \cdot f(x_{i+1}) + d \cdot f(x_{i+2})$$

$$= a \cdot \left[f(x_i) + (-h) \cdot f'(x_i) + \frac{(-h)^2}{2!} \cdot f''(x_i) + \frac{(-h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ b \cdot f(x_i)$$

$$+ c \cdot \left[f(x_i) + (h) \cdot f'(x_i) + \frac{(h)^2}{2!} \cdot f''(x_i) + \frac{(h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$+ d \cdot \left[f(x_i) + (2h) \cdot f'(x_i) + \frac{(2h)^2}{2!} \cdot f''(x_i) + \frac{(2h)^3}{3!} \cdot f^{(3)}(x_i) + \dots \right]$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 1 & 2 \\ (-1)^2 & 0 & (1)^2 & (2)^2 \\ (-1)^3 & 0 & (1)^3 & (2)^3 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{6}{h^3} \end{pmatrix}$$

$$f^{(3)}(x_i) \approx \frac{-f(x_{i-1}) + 3 \cdot f(x_i) - 3 \cdot f(x_{i+1}) + f(x_{i+2})}{h^3}$$

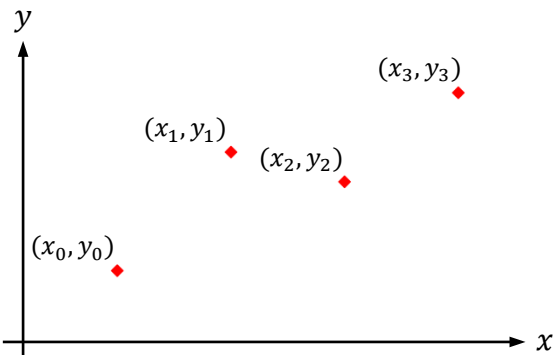
$$f^{(3)}(0.6) \approx \frac{-(-4.827866) + 3 \cdot (-4.240058) - 3 \cdot (-3.496909) + (-2.596792)}{(0.1)^3}$$

$$a = -\frac{1}{h^3}, b = \frac{3}{h^3}, c = -\frac{3}{h^3}, d = \frac{1}{h^3}$$

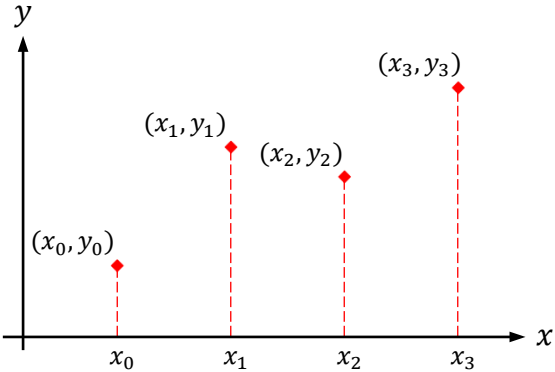
$$= 1.627000$$

The Method of Undetermined Coefficients

The Method of Undetermined Coefficients

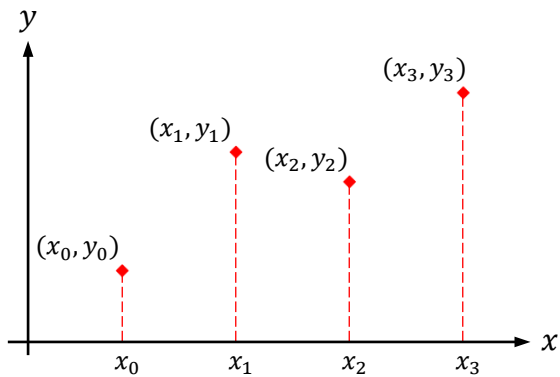


The Method of Undetermined Coefficients



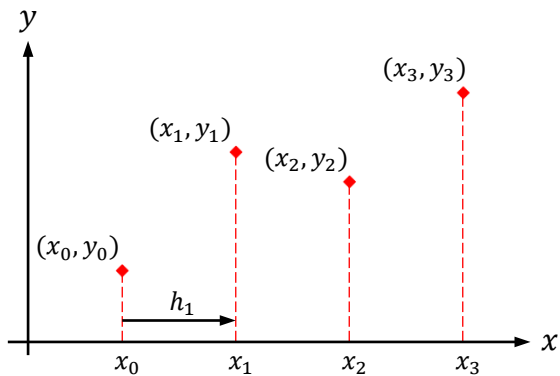
The Method of Undetermined Coefficients

Formulas for Non-Uniform Intervals



The Method of Undetermined Coefficients

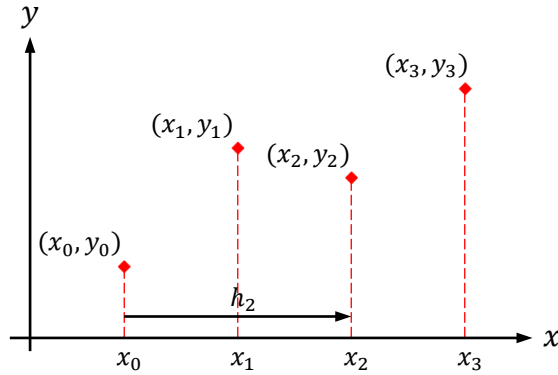
Formulas for Non-Uniform Intervals



$$x_1 - x_0 = h_1$$

The Method of Undetermined Coefficients

Formulas for Non-Uniform Intervals

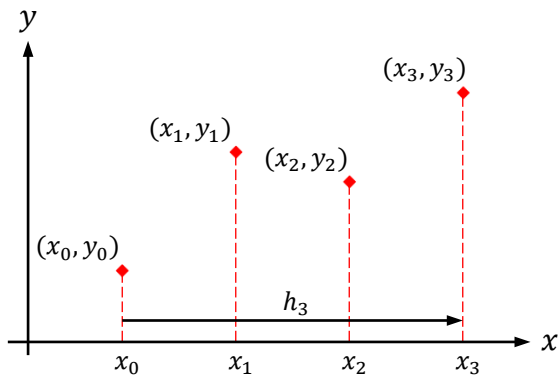


$$x_1 - x_0 = h_1$$

$$x_2 - x_0 = h_2$$

The Method of Undetermined Coefficients

Formulas for Non-Uniform Intervals



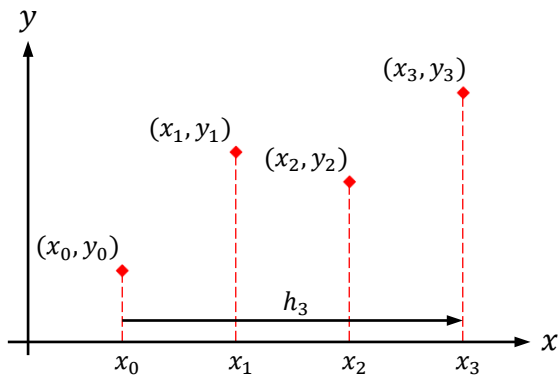
$$x_1 - x_0 = h_1$$

$$x_2 - x_0 = h_2$$

$$x_3 - x_0 = h_3$$

The Method of Undetermined Coefficients

Formulas for Non-Uniform Intervals



$$x_1 - x_0 = h_1$$

$$x_2 - x_0 = h_2$$

$$x_3 - x_0 = h_3$$

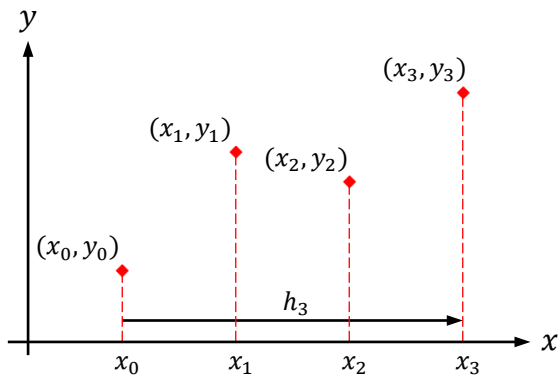
$$\vdots$$

$$x_i - x_0 = h_i$$

The Method of Undetermined Coefficients

Formulas for Non-Uniform Intervals

If data at $n + 1$ points is provided, $(x_i, f(x_i)) \mid i = 0, 1, 2, \dots, n$; k^{th} derivative of $f(x)$ at x_0 when $1 \leq k \leq n$ can be approximated by the following expression.



$$x_1 - x_0 = h_1$$

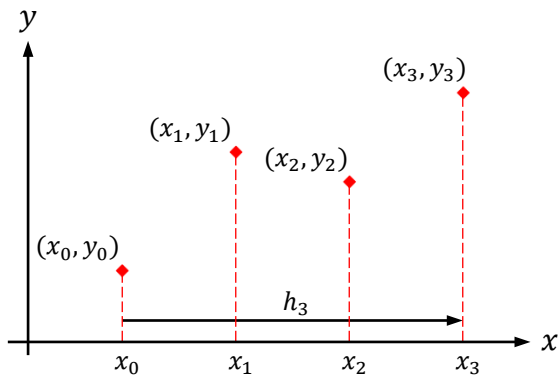
$$x_2 - x_0 = h_2$$

$$x_3 - x_0 = h_3$$

$$\vdots$$

$$x_i - x_0 = h_i$$

The Method of Undetermined Coefficients



$$x_1 - x_0 = h_1$$

$$x_2 - x_0 = h_2$$

$$x_3 - x_0 = h_3$$

$$\vdots$$

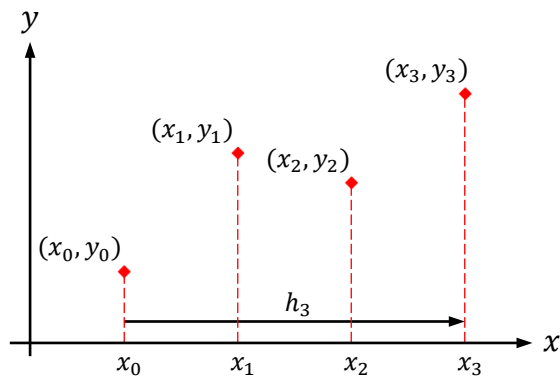
$$x_i - x_0 = h_i$$

Formulas for Non-Uniform Intervals

If data at $n + 1$ points is provided, $(x_i, f(x_i)) \mid i = 0, 1, 2, \dots, n$; k^{th} derivative of $f(x)$ at x_0 when $1 \leq k \leq n$ can be approximated by the following expression.

$$\sum_{i=0}^n w_i \cdot f(x_i) = w_0 \cdot f(x_0) + w_1 \cdot f(x_1) + \dots + w_n \cdot f(x_n); \quad n \geq 1$$

The Method of Undetermined Coefficients



$$x_1 - x_0 = h_1$$

$$x_2 - x_0 = h_2$$

$$x_3 - x_0 = h_3$$

$\vdots$

$$x_i - x_0 = h_i$$

Formulas for Non-Uniform Intervals

If data at $n + 1$ points is provided, $(x_i, f(x_i)) \mid i = 0, 1, 2, \dots, n$; k^{th} derivative of $f(x)$ at x_0 when $1 \leq k \leq n$ can be approximated by the following expression.

$$\sum_{i=0}^n w_i \cdot f(x_i) = w_0 \cdot f(x_0) + w_1 \cdot f(x_1) + \dots + w_n \cdot f(x_n); \quad n \geq 1$$

Taylor series expanded around x_0 are substituted into the equation, then coefficients of the first $n + 1$ terms, including k^{th} derivative, are collected to set up system of equations to find all weighting factors.

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

x_i	$f(x_i)$
0.4	-4.827866
0.6	-4.240058
0.7	-3.496909
0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

x_i	$f(x_i)$
0.4	-4.827866
0.6	-4.240058
0.7	-3.496909
0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

i	x_i	$f(x_i)$
1	0.4	-4.827866
0	0.6	-4.240058
2	0.7	-3.496909
3	0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

i	x_i	$f(x_i)$	h_i
1	0.4	-4.827866	-0.2
0	0.6	-4.240058	-
2	0.7	-3.496909	0.1
3	0.8	-2.596792	0.2

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

i	x_i	$f(x_i)$	h_i
1	0.4	-4.827866	-0.2
0	0.6	-4.240058	-
2	0.7	-3.496909	0.1
3	0.8	-2.596792	0.2

$$a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3)$$

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

i	x_i	$f(x_i)$	h_i
1	0.4	-4.827866	-0.2
0	0.6	-4.240058	-
2	0.7	-3.496909	0.1
3	0.8	-2.596792	0.2

$$a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3)$$

$$= a \cdot f(x_0)$$

$$+ b \cdot \left[f(x_0) + (h_1) \cdot f'(x_0) + \frac{(h_1)^2}{2!} \cdot f''(x_0) + \frac{(h_1)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right]$$

$$+ c \cdot \left[f(x_0) + (h_2) \cdot f'(x_0) + \frac{(h_2)^2}{2!} \cdot f''(x_0) + \frac{(h_2)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right]$$

$$+ d \cdot \left[f(x_0) + (h_3) \cdot f'(x_0) + \frac{(h_3)^2}{2!} \cdot f''(x_0) + \frac{(h_3)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right]$$

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

i	x_i	$f(x_i)$	h_i
1	0.4	-4.827866	-0.2
0	0.6	-4.240058	-
2	0.7	-3.496909	0.1
3	0.8	-2.596792	0.2

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & h_1 & h_2 & h_3 \\ 0 & h_1^2 & h_2^2 & h_3^2 \\ 0 & h_1^3 & h_2^3 & h_3^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$$

$$a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3)$$

$$= a \cdot f(x_0)$$

$$+ b \cdot \left[f(x_0) + (h_1) \cdot f'(x_0) + \frac{(h_1)^2}{2!} \cdot f''(x_0) + \frac{(h_1)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right]$$

$$+ c \cdot \left[f(x_0) + (h_2) \cdot f'(x_0) + \frac{(h_2)^2}{2!} \cdot f''(x_0) + \frac{(h_2)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right]$$

$$+ d \cdot \left[f(x_0) + (h_3) \cdot f'(x_0) + \frac{(h_3)^2}{2!} \cdot f''(x_0) + \frac{(h_3)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right]$$

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

i	x_i	$f(x_i)$	h_i
1	0.4	-4.827866	-0.2
0	0.6	-4.240058	-
2	0.7	-3.496909	0.1
3	0.8	-2.596792	0.2

$$\begin{aligned} &a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3) \\ &= a \cdot f(x_0) \\ &+ b \cdot \left[f(x_0) + (h_1) \cdot f'(x_0) + \frac{(h_1)^2}{2!} \cdot f''(x_0) + \frac{(h_1)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right] \\ &+ c \cdot \left[f(x_0) + (h_2) \cdot f'(x_0) + \frac{(h_2)^2}{2!} \cdot f''(x_0) + \frac{(h_2)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right] \\ &+ d \cdot \left[f(x_0) + (h_3) \cdot f'(x_0) + \frac{(h_3)^2}{2!} \cdot f''(x_0) + \frac{(h_3)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right] \end{aligned}$$

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & h_1 & h_2 & h_3 \\ 0 & h_1^2 & h_2^2 & h_3^2 \\ 0 & h_1^3 & h_2^3 & h_3^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$$

$$a = -\frac{1}{h_1} - \frac{1}{h_2} - \frac{1}{h_3} = -10$$

$$b = \frac{h_2 h_3}{h_1 \cdot (h_1 - h_2) \cdot (h_1 - h_3)} = -0.833333$$

$$c = \frac{h_1 h_3}{h_2 \cdot (h_2 - h_1) \cdot (h_2 - h_3)} = 13.333333$$

$$d = \frac{h_1 h_2}{h_3 \cdot (h_3 - h_1) \cdot (h_3 - h_2)} = -2.5$$

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

i	x_i	$f(x_i)$	h_i
1	0.4	-4.827866	-0.2
0	0.6	-4.240058	-
2	0.7	-3.496909	0.1
3	0.8	-2.596792	0.2

$$\begin{aligned} &a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3) \\ &= a \cdot f(x_0) \\ &+ b \cdot \left[f(x_0) + (h_1) \cdot f'(x_0) + \frac{(h_1)^2}{2!} \cdot f''(x_0) + \frac{(h_1)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right] \\ &+ c \cdot \left[f(x_0) + (h_2) \cdot f'(x_0) + \frac{(h_2)^2}{2!} \cdot f''(x_0) + \frac{(h_2)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right] \\ &+ d \cdot \left[f(x_0) + (h_3) \cdot f'(x_0) + \frac{(h_3)^2}{2!} \cdot f''(x_0) + \frac{(h_3)^3}{3!} \cdot f^{(3)}(x_0) + \dots \right] \end{aligned}$$

$f(x_i)$
 $f'(x_i)$
 $f''(x_i)$
 $f^{(3)}(x_i)$

$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & h_1 & h_2 & h_3 \\ 0 & h_1^2 & h_2^2 & h_3^2 \\ 0 & h_1^3 & h_2^3 & h_3^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$

$a = -\frac{1}{h_1} - \frac{1}{h_2} - \frac{1}{h_3} = -10$
 $b = \frac{h_2 h_3}{h_1 \cdot (h_1 - h_2) \cdot (h_1 - h_3)} = -0.833333$

$c = \frac{h_1 h_3}{h_2 \cdot (h_2 - h_1) \cdot (h_2 - h_3)} = 13.333333$
 $d = \frac{h_1 h_2}{h_3 \cdot (h_3 - h_1) \cdot (h_3 - h_2)} = -2.5$

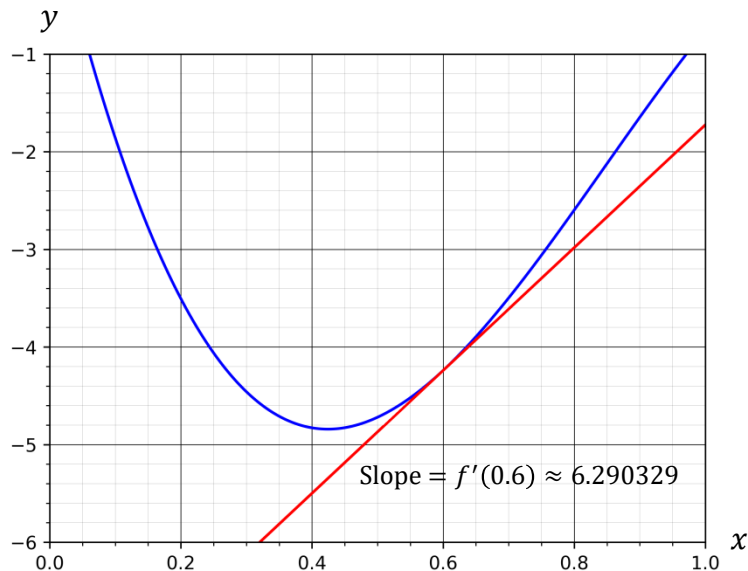
$f'(0.6) \approx -0.833333 \cdot (-4.827866) - 10 \cdot (-4.240058) + 13.333333 \cdot (-3.496909) - 2.5 \cdot (-2.596792) = 6.290329$

The Method of Undetermined Coefficients

Example 05-05: Approximate the value of $f'(0.6)$ from the following data.

i	x_i	$f(x_i)$	h_i
1	0.4	-4.827866	-0.2
0	0.6	-4.240058	—
2	0.7	-3.496909	0.1
3	0.8	-2.596792	0.2

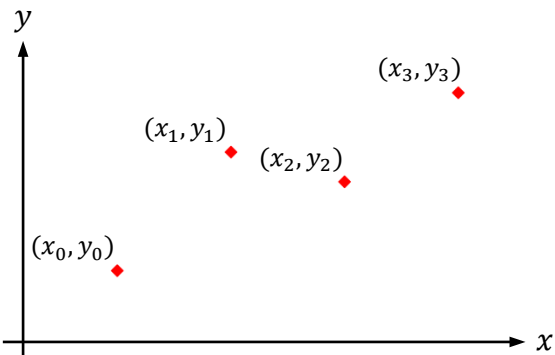
$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & h_1 & h_2 & h_3 \\ 0 & h_1^2 & h_2^2 & h_3^2 \\ 0 & h_1^3 & h_2^3 & h_3^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$$



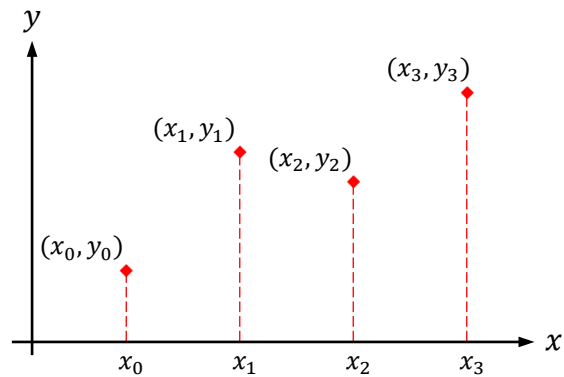
$$f'(0.6) \approx -0.833333 \cdot (-4.827866) - 10 \cdot (-4.240058) + 13.333333 \cdot (-3.496909) - 2.5 \cdot (-2.596792) = 6.290329$$

The Method of Undetermined Coefficients

The Method of Undetermined Coefficients

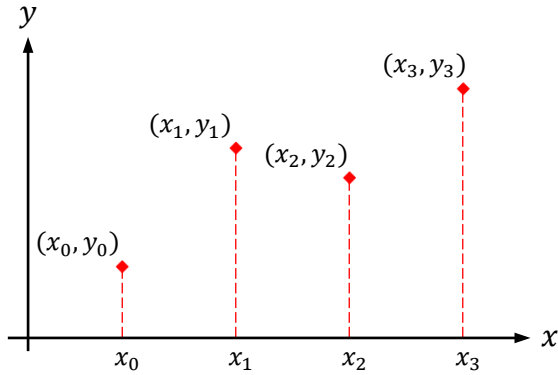


The Method of Undetermined Coefficients



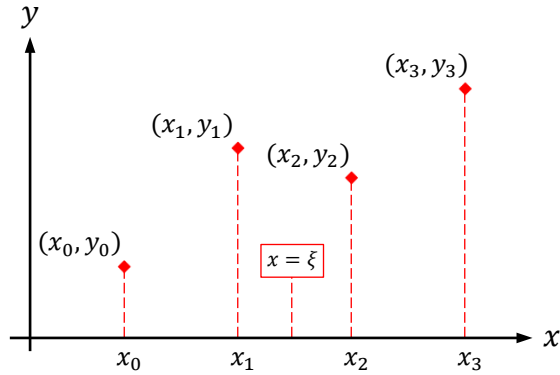
The Method of Undetermined Coefficients

Formulas at Arbitrary Points



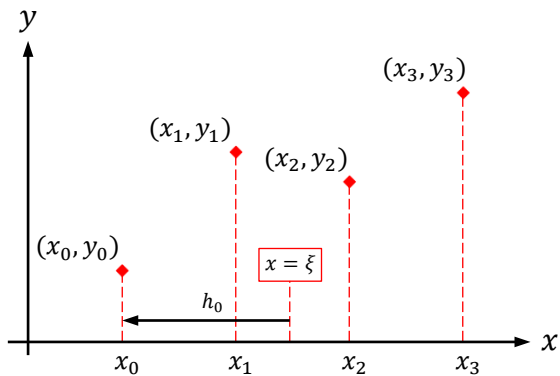
The Method of Undetermined Coefficients

Formulas at Arbitrary Points



The Method of Undetermined Coefficients

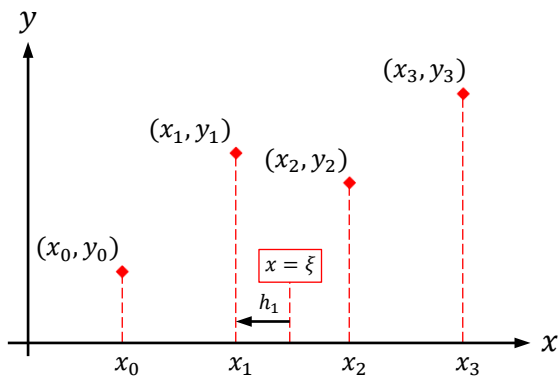
Formulas at Arbitrary Points



$$x_0 - \xi = h_0$$

The Method of Undetermined Coefficients

Formulas at Arbitrary Points

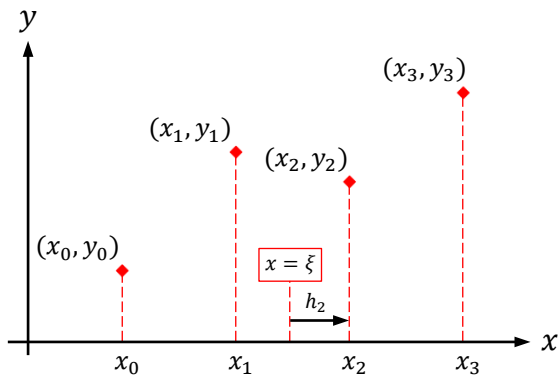


$$x_0 - \xi = h_0$$

$$x_1 - \xi = h_1$$

The Method of Undetermined Coefficients

Formulas at Arbitrary Points



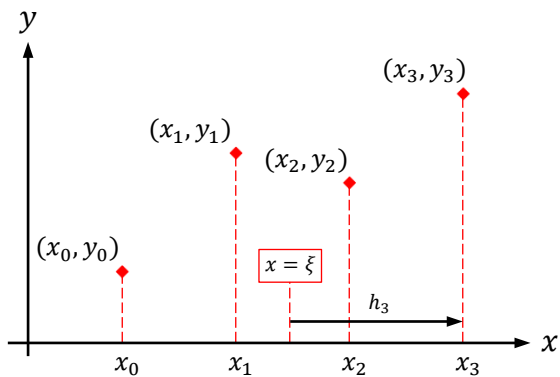
$$x_0 - \xi = h_0$$

$$x_1 - \xi = h_1$$

$$x_2 - \xi = h_2$$

The Method of Undetermined Coefficients

Formulas at Arbitrary Points



$$x_0 - \xi = h_0$$

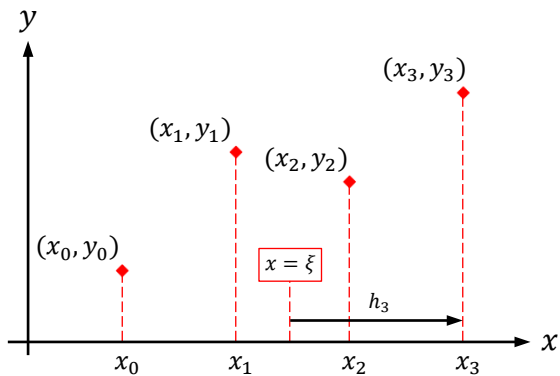
$$x_1 - \xi = h_1$$

$$x_2 - \xi = h_2$$

$$x_3 - \xi = h_3$$

The Method of Undetermined Coefficients

Formulas at Arbitrary Points



$$x_0 - \xi = h_0$$

$$x_1 - \xi = h_1$$

$$x_2 - \xi = h_2$$

$$x_3 - \xi = h_3$$

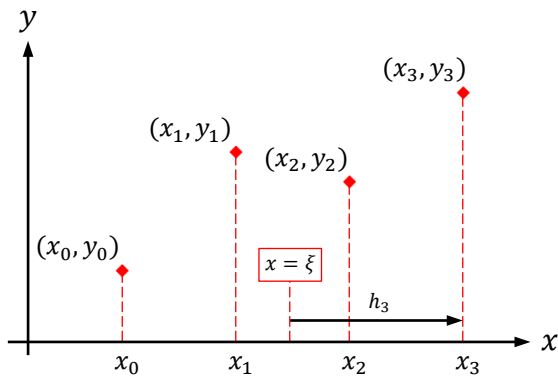
$$\vdots$$

$$x_i - \xi = h_i$$

The Method of Undetermined Coefficients

Formulas at Arbitrary Points

If data at $n + 1$ points is provided, $(x_i, f(x_i)) \mid i = 0, 1, 2, \dots, n$;
 k^{th} derivative of $f(x)$ at ξ when $1 \leq k \leq n$ can be approximated
by the following expression.



$$x_0 - \xi = h_0$$

$$x_1 - \xi = h_1$$

$$x_2 - \xi = h_2$$

$$x_3 - \xi = h_3$$

$$\vdots$$

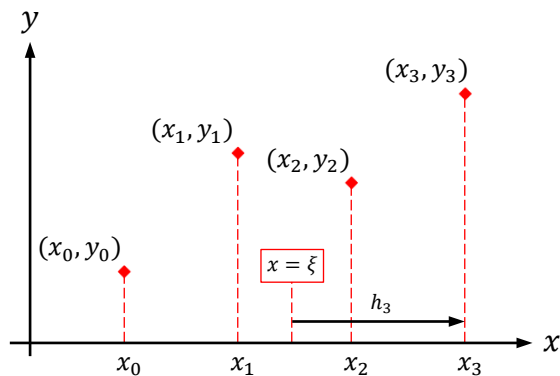
$$x_i - \xi = h_i$$

The Method of Undetermined Coefficients

Formulas at Arbitrary Points

If data at $n + 1$ points is provided, $(x_i, f(x_i)) \mid i = 0, 1, 2, \dots, n$; k^{th} derivative of $f(x)$ at ξ when $1 \leq k \leq n$ can be approximated by the following expression.

$$\sum_{i=0}^n w_i \cdot f(x_i) = w_0 \cdot f(x_0) + w_1 \cdot f(x_1) + \dots + w_n \cdot f(x_n); \quad n \geq 1$$



$$x_0 - \xi = h_0$$

$$x_1 - \xi = h_1$$

$$x_2 - \xi = h_2$$

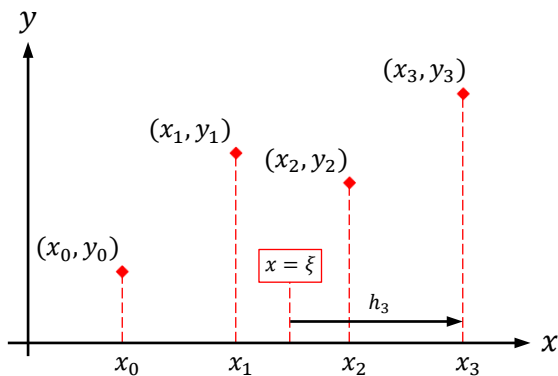
$$x_3 - \xi = h_3$$

$$\vdots$$

$$x_i - \xi = h_i$$

The Method of Undetermined Coefficients

Formulas at Arbitrary Points



$$x_0 - \xi = h_0$$

$$x_1 - \xi = h_1$$

$$x_2 - \xi = h_2$$

$$x_3 - \xi = h_3$$

$$\vdots$$

$$x_i - \xi = h_i$$

If data at $n + 1$ points is provided, $(x_i, f(x_i)) \mid i = 0, 1, 2, \dots, n$; k^{th} derivative of $f(x)$ at ξ when $1 \leq k \leq n$ can be approximated by the following expression.

$$\sum_{i=0}^n w_i \cdot f(x_i) = w_0 \cdot f(x_0) + w_1 \cdot f(x_1) + \dots + w_n \cdot f(x_n); \quad n \geq 1$$

Taylor series expanded around ξ are substituted into the equation, then coefficients of the first $n + 1$ terms, including k^{th} derivative, are collected to set up system of equations to find all weighting factors.

The Method of Undetermined Coefficients

Example 05-06: Approximate the value of $f'(0.5)$ from the following data.

x_i	$f(x_i)$
0.4	-4.827866
0.6	-4.240058
0.7	-3.496909
0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-06: Approximate the value of $f'(0.5)$ from the following data.

i	x_i	$f(x_i)$
0	0.4	-4.827866
1	0.6	-4.240058
2	0.7	-3.496909
3	0.8	-2.596792

The Method of Undetermined Coefficients

Example 05-06: Approximate the value of $f'(0.5)$ from the following data.

i	x_i	$f(x_i)$	h_i
0	0.4	-4.827866	-0.1
1	0.6	-4.240058	+0.1
2	0.7	-3.496909	+0.2
3	0.8	-2.596792	+0.3

The Method of Undetermined Coefficients

Example 05-06: Approximate the value of $f'(0.5)$ from the following data.

<i>i</i>	x_i	$f(x_i)$	h_i
0	0.4	-4.827866	-0.1
1	0.6	-4.240058	+0.1
2	0.7	-3.496909	+0.2
3	0.8	-2.596792	+0.3

$$a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3)$$

The Method of Undetermined Coefficients

Example 05-06: Approximate the value of $f'(0.5)$ from the following data.

i	x_i	$f(x_i)$	h_i
0	0.4	-4.827866	-0.1
1	0.6	-4.240058	+0.1
2	0.7	-3.496909	+0.2
3	0.8	-2.596792	+0.3

$$\begin{aligned} & a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3) \\ &= a \cdot \left[f(\xi) + (h_0) \cdot f'(\xi) + \frac{(h_0)^2}{2!} \cdot f''(\xi) + \frac{(h_0)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right] \\ &+ b \cdot \left[f(\xi) + (h_1) \cdot f'(\xi) + \frac{(h_1)^2}{2!} \cdot f''(\xi) + \frac{(h_1)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right] \\ &+ c \cdot \left[f(\xi) + (h_2) \cdot f'(\xi) + \frac{(h_2)^2}{2!} \cdot f''(\xi) + \frac{(h_2)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right] \\ &+ d \cdot \left[f(\xi) + (h_3) \cdot f'(\xi) + \frac{(h_3)^2}{2!} \cdot f''(\xi) + \frac{(h_3)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right] \end{aligned}$$

The Method of Undetermined Coefficients

Example 05-06: Approximate the value of $f'(0.5)$ from the following data.

i	x_i	$f(x_i)$	h_i
0	0.4	-4.827866	-0.1
1	0.6	-4.240058	+0.1
2	0.7	-3.496909	+0.2
3	0.8	-2.596792	+0.3

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ h_0 & h_1 & h_2 & h_3 \\ h_0^2 & h_1^2 & h_2^2 & h_3^2 \\ h_0^3 & h_1^3 & h_2^3 & h_3^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$$

$$\begin{aligned} & a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3) \\ &= a \cdot \left[f(\xi) + (h_0) \cdot f'(\xi) + \frac{(h_0)^2}{2!} \cdot f''(\xi) + \frac{(h_0)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right] \\ &+ b \cdot \left[f(\xi) + (h_1) \cdot f'(\xi) + \frac{(h_1)^2}{2!} \cdot f''(\xi) + \frac{(h_1)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right] \\ &+ c \cdot \left[f(\xi) + (h_2) \cdot f'(\xi) + \frac{(h_2)^2}{2!} \cdot f''(\xi) + \frac{(h_2)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right] \\ &+ d \cdot \left[f(\xi) + (h_3) \cdot f'(\xi) + \frac{(h_3)^2}{2!} \cdot f''(\xi) + \frac{(h_3)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right] \end{aligned}$$

The Method of Undetermined Coefficients

Example 05-06: Approximate the value of $f'(0.5)$ from the following data.

i	x_i	$f(x_i)$	h_i
0	0.4	-4.827866	-0.1
1	0.6	-4.240058	+0.1
2	0.7	-3.496909	+0.2
3	0.8	-2.596792	+0.3

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ h_0 & h_1 & h_2 & h_3 \\ h_0^2 & h_1^2 & h_2^2 & h_3^2 \\ h_0^3 & h_1^3 & h_2^3 & h_3^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$$

$$a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3)$$

$$= a \cdot \left[f(\xi) + (h_0) \cdot f'(\xi) + \frac{(h_0)^2}{2!} \cdot f''(\xi) + \frac{(h_0)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right]$$

$$+ b \cdot \left[f(\xi) + (h_1) \cdot f'(\xi) + \frac{(h_1)^2}{2!} \cdot f''(\xi) + \frac{(h_1)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right]$$

$$+ c \cdot \left[f(\xi) + (h_2) \cdot f'(\xi) + \frac{(h_2)^2}{2!} \cdot f''(\xi) + \frac{(h_2)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right]$$

$$+ d \cdot \left[f(\xi) + (h_3) \cdot f'(\xi) + \frac{(h_3)^2}{2!} \cdot f''(\xi) + \frac{(h_3)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right]$$

$$a = \frac{h_1 h_2 + h_1 h_3 + h_2 h_3}{(h_0 - h_1) \cdot (h_0 - h_2) \cdot (h_0 - h_3)}$$

$$b = \frac{h_0 h_2 + h_0 h_3 + h_2 h_3}{(h_1 - h_0) \cdot (h_1 - h_2) \cdot (h_1 - h_3)}$$

$$c = \frac{h_0 h_1 + h_0 h_3 + h_1 h_3}{(h_2 - h_0) \cdot (h_2 - h_1) \cdot (h_2 - h_3)}$$

$$d = \frac{h_0 h_1 + h_0 h_2 + h_1 h_2}{(h_3 - h_0) \cdot (h_3 - h_1) \cdot (h_3 - h_2)}$$

The Method of Undetermined Coefficients

Example 05-06: Approximate the value of $f'(0.5)$ from the following data.

i	x_i	$f(x_i)$	h_i
0	0.4	-4.827866	-0.1
1	0.6	-4.240058	+0.1
2	0.7	-3.496909	+0.2
3	0.8	-2.596792	+0.3

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ h_0 & h_1 & h_2 & h_3 \\ h_0^2 & h_1^2 & h_2^2 & h_3^2 \\ h_0^3 & h_1^3 & h_2^3 & h_3^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$$

$$a \cdot f(x_0) + b \cdot f(x_1) + c \cdot f(x_2) + d \cdot f(x_3)$$

$$= a \cdot \left[f(\xi) + (h_0) \cdot f'(\xi) + \frac{(h_0)^2}{2!} \cdot f''(\xi) + \frac{(h_0)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right]$$

$$+ b \cdot \left[f(\xi) + (h_1) \cdot f'(\xi) + \frac{(h_1)^2}{2!} \cdot f''(\xi) + \frac{(h_1)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right]$$

$$+ c \cdot \left[f(\xi) + (h_2) \cdot f'(\xi) + \frac{(h_2)^2}{2!} \cdot f''(\xi) + \frac{(h_2)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right]$$

$$+ d \cdot \left[f(\xi) + (h_3) \cdot f'(\xi) + \frac{(h_3)^2}{2!} \cdot f''(\xi) + \frac{(h_3)^3}{3!} \cdot f^{(3)}(\xi) + \dots \right]$$

$$a = \frac{h_1 h_2 + h_1 h_3 + h_2 h_3}{(h_0 - h_1) \cdot (h_0 - h_2) \cdot (h_0 - h_3)}$$

$$b = \frac{h_0 h_2 + h_0 h_3 + h_2 h_3}{(h_1 - h_0) \cdot (h_1 - h_2) \cdot (h_1 - h_3)}$$

$$c = \frac{h_0 h_1 + h_0 h_3 + h_1 h_3}{(h_2 - h_0) \cdot (h_2 - h_1) \cdot (h_2 - h_3)}$$

$$d = \frac{h_0 h_1 + h_0 h_2 + h_1 h_2}{(h_3 - h_0) \cdot (h_3 - h_1) \cdot (h_3 - h_2)}$$

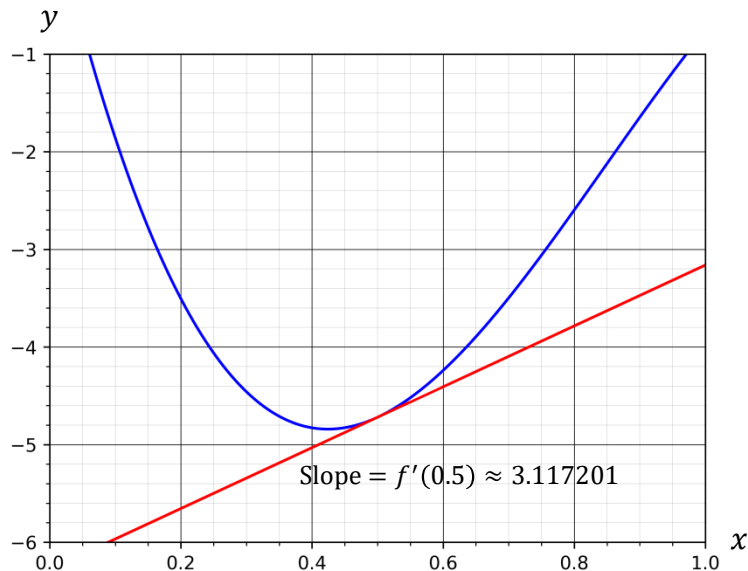
$$f'(0.5) \approx -4.58333 \cdot (-4.827866) + 2.500000 \cdot (-4.240058) + 3.000000 \cdot (-3.496909) - 1.250000 \cdot (-2.596792) = 3.117201$$

The Method of Undetermined Coefficients

Example 05-06: Approximate the value of $f'(0.5)$ from the following data.

i	x_i	$f(x_i)$	h_i
0	0.4	-4.827866	-0.1
1	0.6	-4.240058	+0.1
2	0.7	-3.496909	+0.2
3	0.8	-2.596792	+0.3

$$\begin{matrix} f(x_i) \\ f'(x_i) \\ f''(x_i) \\ f^{(3)}(x_i) \end{matrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ h_0 & h_1 & h_2 & h_3 \\ h_0^2 & h_1^2 & h_2^2 & h_3^2 \\ h_0^3 & h_1^3 & h_2^3 & h_3^3 \end{bmatrix} \begin{Bmatrix} a \\ b \\ c \\ d \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{Bmatrix}$$



$$f'(0.5) \approx -4.58333 \cdot (-4.827866) + 2.500000 \cdot (-4.240058) + 3.000000 \cdot (-3.496909) - 1.250000 \cdot (-2.596792) = 3.117201$$